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Feature-based magnetotelluric inversion by variational autoencoder using a subdomain encoding scheme

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ABSTRACT

Magnetotelluric (MT) data inversion aims to reconstruct a subsurface resistivity model that minimizes the discrepancy between inverted and measured electromagnetic data. Conventional pixel-based minimum-structure inversion often yields a smoothed-out reconstruction with a relatively low resolution. A priori geophysical knowledge can be embedded into inversion and improve the reconstruction resolution through proper reparameterization. However, existing reparameterization approaches, such as model-based and parametric transform-based inversion, have limited ability to incorporate various a priori information. The effectiveness of existing deep generative model-based inversion algorithms is still debatable when applied to scenarios with complex backgrounds. We develop a feature-based MT data inversion method based on a variational autoencoder (VAE) with a subdomain encoding scheme. Instead of encoding the entire domain of an investigation, we adopt a 1D subdomain encoding scheme to encode the 1D resistivity-depth models using a single VAE. The latent variables for the 2D model are a combination of the latent variables for 1D models, and the encoded region of interest (ROI) can be flexibly determined. The latent variables of ROI and the pixels outside the ROI are simultaneously inverted using the gradient-descent method. Our 1D subdomain encoding scheme reduces the complexity and diversity of the data set, and it can flexibly embed a priori knowledge with various uncertainties. Synthetic data inversion and inversion of the Southern African Magnetotelluric Experiment field data validate our method's ability to effectively improve inversion accuracy and resolution.

INTRODUCTION

Magnetotelluric (MT) data inversion aims to reconstruct a resistivity model of the subsurface by minimizing the difference between the inverted and measured data. To achieve this, the continuous geophysical model needs to be appropriately parameterized. One commonly used parameterization method is to discretize the subsurface model into pixels assigned with material values, which are the unknowns of the inversion. In deterministic approaches, the objective function typically consists of the data misfit between inverted and measured data, as well as regularization, such as the Tikhonov regularization based on L_1- or L_2-norm smoothness. This function is often minimized using iterative optimization techniques, such as gradient- or Newton-based methods. This parameterization method is widely studied and used in earlier works, such as Constable et al. (1987), Oldenburg and Li (1994), Loke and Barker (1996), Habashy and Abubakar (2004), Abubakar et al. (2008), Commer and Newman (2008), Key (2009), and others. Although pixel-based parameterization is straightforward and little dependent on a priori knowledge, it has two significant drawbacks. First, it can lead to millions of unknowns in 3D inversion, making computation costs considerable. Second, incorporating a priori knowledge of the survey area, such as information provided by other prospecting methods or the model's geometry and pattern, is challenging. To address this issue, many reparameterization approaches, such as model-based and parametric transform-based methods, as well as deep-learning methods, have been studied.

The model-based approach uses horizons and closed polygons to describe the blocks and layers in the subsurface model (Li et al., 2010a). It assumes that the material attributes of the layers and blocks are constant and invert the central coordinates, node coordinates, and material attribute values of the layers and blocks

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(Marcuello-Pascual et al., 1992; Auken and Christiansen, 2004; Li et al., 2010a, 2010b; Chen et al., 2012). This approach reduces the number of parameters, ensures flexibility in the topological structure, and incorporates a priori knowledge through a proper initial model (Li et al., 2010a). However, the model-based approach has high ill-posedness and nonlinearity, and improper initial models can easily lead to local minima.

Parametric transforms can help reduce the number of parameters, alleviate the ill-posed nature of inverse problems, and produce reconstructions with reasonable patterns (Khaninezhad et al., 2012; Vo and Durlofsky, 2016). Classical dimension reduction methods, such as principle component analysis and singular value decomposition, have been used in geologic flow-estimation problems (Sarma et al., 2008; Khaninezhad et al., 2012; Vo and Durlofsky, 2014), acoustic and elastic inversion (Grana et al., 2019), and cross-borehole electrical resistivity tomography data inversion (Oware et al., 2019). Such inversions typically involve generating model realizations with geostatistical knowledge, a linear or nonlinear parametric transform to construct a basis matrix with the data matrix, and formulating an optimization problem with unknowns in transform space. Fourier transform and wavelet transform have been widely studied in many geophysical inverse problems such as MT (Nittinger and Becken, 2016), airborne electromagnetic (EM) (Liu et al., 2018), controlled-source EM (CSEM) (Abubakar et al., 2012), seismic first-arrival traveltime (Delost et al., 2008), and seismic tomography (Chiao and Liang, 2003; Fang and Zhang, 2014). The a priori information on the scale and resolution of the model can be embedded by flexibly selecting wavelet basis functions with different coordinates and scale factors. This approach allows the fine structure of small-scale formations in geophysical models to be more clearly characterized, resulting in a multiresolution reconstruction based on a priori knowledge. However, it is challenging to flexibly embed more ad hoc knowledge, such as specific model patterns or numerical constraints on material attributes.

With recent advancements in big data science, parallel computing, and computational optimization, machine-learning techniques have been extensively used in various forms to solve inverse problems (Huang et al., 2021; Zhang and Curtis, 2021; Cai et al., 2022). In one approach, problem-specific training data are not mandatory. For instance, in Jin et al. (2020) and Sun and Alkhalifah (2020), a recurrent neural network is used to construct an iterative physical equation solver and predict the optimization gradient through data residuals and historical descent direction embedded in hidden states. Some studies construct a pseudoinverse operator with the neural network and adopt the idea of progressive training (Alyousuf and Li, 2021; Colombo et al., 2021). The output of the neural network provides a data-driven regularization for the inversion, and the network is optimized with current data and model as the inversion proceeds. This approach often uses neural networks to replace a process or calculation unit in inverse problems, and the requirement for the size and specificity of the data set is relatively low.

The machine-learning approach also allows for the incorporation of more ad hoc a priori knowledge. This knowledge can be in the form of multiphysics data and models, or experts' expertise in the research area, which can provide additional information to improve inversion accuracy and resolution (Tarantola and Valette, 1982). For example, salt delineation with CSEM can be improved by seismic data using a convolutional neural network (CNN) (Oh et al., 2020). In a 3D sparse inversion of gravity data, a complex and diverse density model training set is generated according to the geometric pattern of the subsurface density distribution (Huang et al., 2021). Deep-learning-based methods are applied to reservoir fluid saturation monitoring, and the influence of a priori information on inversion is discussed (Colombo et al., 2020). The supervised descent method can learn the specified descent direction by adopting a more specified data set (Guo et al., 2020). Di et al. (2021) construct an intelligent earthquake interpretation framework based on CNN, with the constraint of a priori geologic knowledge and solid interpretation expertise of interpreters. With a properly designed data set, machine-learning models can automatically mine the implicit knowledge and incorporate it into the inversion, which can be difficult using methods based on explicit mathematical equations. However, this approach relies on properly designed training sets to contain all practical geophysical model patterns; otherwise, the inversion results may not sufficiently represent the real scenario (Kim and Nakata, 2018; Colombo et al., 2020).

Recently, a deep-learning-based model reparameterization approach has drawn increasing attention (Laloy et al., 2017; Canchumuni et al., 2019; Guo et al., 2022; Lin et al., 2022). Deep generative models (DGMs), e.g., variational autoencoders (VAEs) or generative adversarial networks, can learn low-dimensional representation in a self-supervised manner. DGM-based methods can extract a priori knowledge, such as the distribution of physical attributes and the geometry of subsurface structures, and constrain the inversion with such knowledge (Richardson, 2018). Liu et al. (2022) and Sun and Alkhalifah (2020) adopt VAEs to reduce model dimensionality in underground water flow reconstruction and seismic FWI and accelerate the inversion with less computational cost. McAliley and Li (2021) use the VAE to invert gravity data in a stochastic manner using a random sampling scheme. In ground-penetrating radar traveltime data inversion, Lopez-Alvis et al. (2022) adopt the VAE to assemble a diverse set of a priori patterns and produce more realistic model reconstructions. Lopez-Alvis et al. (2021) study the nonlinearity of DGM-based inversion and present the advantage of the stochastic gradient-descent method over gradient-descent methods.

One essential requirement for the DGM-based inversion is that the training data set shall contain the various patterns with which the model may appear. The existing applications mainly encode the entire research area in the inversion (i.e., the domain of investigation [DOI]) with a single VAE. This approach is relatively straightforward, but it has limited capability in flexibly incorporating a priori knowledge with different characteristics, such as a priori knowledge with complex patterns, high uncertainty, and limited spatial scope. Increasing the size and diversity of the training set can alleviate this problem, but this imposes high requirements on the training set and leads to extended training time. For now, data inversion under complex backgrounds (e.g., the geophysical EM inverse problem) with VAEs still faces many obstacles.

In this work, we study a feature-based 2D MT data inversion method using a VAE with a subdomain encoding scheme. Instead of encoding the DOI as a whole, we adopt a 1D subdomain encoding scheme to, respectively, encode the 1D resistivity-depth models with a single VAE. The inversion unknowns include the latent variables of the interested subregion inside DOI (i.e., region of interest [ROI]) and the pixels outside the ROI, and the latent variables of 2D ROI are the combination of latent variables for the 1D models (the ranges of the DOI and ROI are shown in Figure 1). Adopting the chain derivation rule and the deep-learning automatic differentiation (AD) framework, the latent variables and pixels are inverted using gradient-descent algorithms. The ROI recovered from the latent variables exhibits a priori patterns, whereas the reconstruction outside the ROI resembles the pixel-based approach. The subdomain encoding scheme can reduce the complexity and diversity requirements of the VAE data set. By reasonably designing the training data set, it is possible to flexibly embed a priori knowledge with various uncertainties and scopes. We apply the proposed algorithm to synthetic experiments and the field inversion of Southern African Magnetotelluric Experiment (SAMTEX) data collected in southern Africa (Jones et al., 2009). The synthetic and field data inversion verifies that the proposed method can improve the accuracy and resolution of model reconstruction.

The remainder of this work is structured as follows. The "Methods" section presents the formation of the feature-based inversion. The "Numerical tests" section presents the results of synthetic experiments, and in the "Application to field data" section, the feature-based inversion is applied to field data inversion. Some discussion of the proposed algorithm is presented in the "Discussion" section. Finally, the conclusion of the work is presented in the "Conclusion" section.

METHODS

Variational autoencoder

The VAE is a descendant of the classical autoencoder (AE) structure. Since its introduction in 2006 (Hinton and Salakhutdinov, 2006), AEs have gained significant attention, leading to extensive research and the development of numerous variants, such as sparse AEs (Ng, 2011), denoising AEs (Vincent et al., 2008), stacked AEs (Vincent et al., 2010), and VAEs (Kingma and Welling, 2013).

An AE defines a neural network that yields a low-dimensional representation of high-dimensional data. In general, it has an hourglass shape in terms of the feature map size, and the most compact feature map in the center is often called the latent variable. The part connecting the input and the latent variable is defined as an encoder, and the part connecting the latent variable and the output is defined as a decoder.

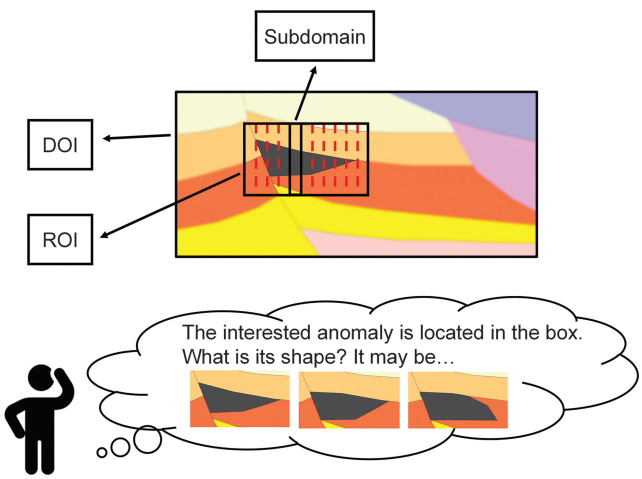

Figure 1. The range of DOI, ROI, and a possible scenario of embedding a priori knowledge in geophysical inversion.

As a generative model, AE requires that the output should approximate the input, and it can generate new models displaying specific patterns after being carefully trained with a proper data set. However, because the AE merely minimizes the discrepancy between the input and output without imposing any constraint on the latent variable distribution, it cannot guarantee a continuous and complete latent space. Latent variables that were not sampled during training may yield implausible models in the test. In the data inversion where the unknowns are the latent variables, a discontinuous and unregularized latent space can lead to an unstable inversion process.

Compared with traditional AEs, VAEs offer the advantage of learning a continuous and structured latent space, which enables more effective generative modeling and sampling of new data points (Kingma and Welling, 2013, 2019; Rezende et al., 2014; Burda et al., 2015; Sønderby et al., 2016). Figure 2 shows the diagram of a typical VAE. We denote the input model with $\mathbf{m}$ and the latent variable with $\mathbf{v}$. Same as the conventional AE structure, the VAE is also composed of two parts, namely an encoder $\mathcal{E}$ and a decoder $\mathcal{D}$. However, with VAEs, the models and latent variables are generated in a stochastic manner. The encoder $\mathcal{E}$ projects the model $\mathbf{m}$ to the latent space with the distribution $q_\phi(\mathbf{v}|\mathbf{m})$ and the decoder $\mathcal{D}$ recovers the model $\mathbf{m}$ from the latent variable $\mathbf{v}$ with the distribution $p_\theta(\mathbf{m}|\mathbf{v})$. In contrast to AE, in VAEs, the parameters of $p_\theta(\mathbf{m}|\mathbf{v})$ and $q_\phi(\mathbf{v}|\mathbf{m})$ are deterministically generated with neural layers and, from these distributions, $\mathbf{m}$ and $\mathbf{v}$ are sampled. Here, ϕ and θ are the learnable weights and biases of the encoder and decoder, respectively. The training of the VAE is self-supervised, in which the sampling of $\mathbf{m}$ and $\mathbf{v}$ is conducted automatically. The training can be summarized by minimizing the evidence lower bound $\mathcal{L}_b(\theta,\phi)$ (Kingma and Welling, 2013, 2019) with optimal network parameters, namely

$$\phi,\theta = \arg\min_{\phi,\theta} \mathcal{L}_b(\theta,\phi)$$

$$= \arg\min_{\phi,\theta} \{\mathcal{L} + D_{\mathrm{KL}}(q_\phi(\mathbf{v}|\mathbf{m}) \| p(\mathbf{v}))\}, \tag{1}$$

where

$$\mathcal{L} = -E_{\mathbf{v}\sim q_\phi(\mathbf{v}|\mathbf{m})}[\log\ p_\theta(\mathbf{m}|\mathbf{v})], \tag{2}$$

$$D_{\mathrm{KL}}(q_\phi(\mathbf{v}|\mathbf{m}) \| p(\mathbf{v})) = E_{\mathbf{v}\sim q_\phi(\mathbf{v}|\mathbf{m})}[\log\ q_\phi(\mathbf{v}|\mathbf{m}) - \log p(\mathbf{v})]. \tag{3}$$

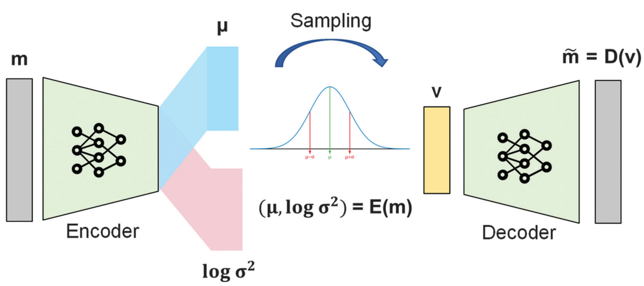

Figure 2. The diagram of a classical VAE structure.

By minimizing equation 1, the VAE simultaneously satisfies the following two constraints imposed by $\mathcal{L}$ and D_{KL}, respectively:

1) The decoder should be able to reconstruct any training sample $\mathbf{m}$ with a maximum posterior probability of $p_\theta(\mathbf{m}|\mathbf{v})$.
2) The posterior of the latent variable $q_\phi(\mathbf{v}|\mathbf{m})$ is refined to approximate the desired distribution $p(\mathbf{v})$ as closely as possible.

The training of the VAE leads to a fine recognition model $p_\theta(\mathbf{m}|\mathbf{v})$ and a fine generative model $q_\phi(\mathbf{v}|\mathbf{m})$ simultaneously. The manifold of the complex model space $\mathcal{M}$ (and hence the generative ability of the VAE) is supported by the latent variable distribution $p(\mathbf{v})$ and model posterior distribution $p_\theta(\mathbf{m}|\mathbf{v})$:

$$p(\mathbf{m}) = \int_{\mathbf{v}} p(\mathbf{v})p_\theta(\mathbf{m}|\mathbf{v}). \tag{4}$$

In practice, these distributions are often assumed to be Gaussian, that is

$$p(\mathbf{v}) \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{5}$$

$$q_\phi(\mathbf{v}|\mathbf{m}) \sim \mathcal{N}(\mu_{\mathcal{E}}, \sigma^2_{\mathcal{E}}), \tag{6}$$

and

$$p_\theta(\mathbf{m}|\mathbf{v}) \sim \mathcal{N}(\mu_{\mathcal{D}}, \sigma^2_{\mathcal{D}}), \tag{7}$$

where $\mathbf{0}$ and $\mathbf{I}$ are the zero vector and identity matrix, respectively. Here, $\mu_{\mathcal{E}}$ and $\sigma^2_{\mathcal{E}}$ are the encoder outputs with

$$(\mu_{\mathcal{E}}, \log\sigma^2_{\mathcal{E}}) = \mathcal{E}(\mathbf{m}) = (\mathcal{E}_1(\mathbf{m}), \mathcal{E}_2(\mathbf{m})), \tag{8}$$

where $\mu_{\mathcal{D}}$ is the decoder output $\mathcal{D}(\mathbf{v})$ and $\sigma^2_{\mathcal{D}}$ is assumed to be a constant matrix $c\mathbf{I}$. Adopting the reparameterization trick (Kingma and Welling, 2013) and after some algebra, the two terms of the loss function become

$$\mathcal{L} = \frac{1}{2c}\|\mathbf{m} - \mathcal{D}(\tilde{\mathbf{v}})\|^2 \tag{9}$$

and

$$D_{\mathrm{KL}} = \frac{1}{2}(\mu^2_{\mathcal{E}} + \sigma^2_{\mathcal{E}} - \log\sigma^2_{\mathcal{E}} - 1), \tag{10}$$

where $\mathbf{m}$ and $\tilde{\mathbf{v}}$ denote the label model and the latent variable sampled from $p(\mathbf{v})$, respectively.

In summary, the VAE establishes a stochastic map between the randomly distributed model $\mathbf{m}$ and the latent variable $\mathbf{v}$ conforming to the standard normal distribution. The remarkable features of VAEs, e.g., the continuous latent space and the competent generative ability, pave the way for their application to inverse problems.

Feature-based inversion with a subdomain encoding scheme

Before introducing the inversion algorithm in detail, we first need to define the concept of a priori knowledge. A priori knowledge often refers to the geophysical or petrophysical understanding of the research area that is obtained before conducting the inversion. The a priori knowledge can have various uncertainties and scopes, and it is to some extent subjective, and even biased. Therefore, it is necessary to comprehensively use both the information from the measured data and the a priori knowledge to ensure the accuracy of the inversion. For example, in Figure 1, the location of the anomaly is known before the inversion, but the shape needs to be determined through the measured data. In the deterministic framework, the prior distribution of the subsurface model can be represented by a set of synthetically generated models, which also serves as the training data set for the neural networks.

Similar to pixel-based inversion, the parametric transform-based inversion can also be solved using deterministic approaches, as detailed in Appendix A. In typical deterministic inversions, a Tikhonov regularization is often added, in addition to the data discrepancy term, to enforce smoothness in the inverted model, such as regularization based on L_2-norm or weighted L_2-norm. Because (see equations A-1–A-6) the model $\mathbf{m}$ is reparameterized by the VAE decoder with

$$\mathbf{m} = \mathcal{D}(\mathbf{v}), \tag{11}$$

the objective function can be summarized as

$$\mathcal{L}(\mathbf{v}) = \|\mathbf{W}_d(\mathbf{d}^{\mathrm{obs}} - \mathcal{F}\circ\mathcal{D}(\mathbf{v}))\|^2 + \alpha\|\mathbf{W}_v(\mathbf{v} - \mathbf{v}_{\mathrm{ref}})\|^2, \tag{12}$$

where $\mathbf{W}_d$ and $\mathbf{W}_v$ are the terms weighting the relative importance of elements, α denotes the trade-off parameter weighting the data discrepancy and regularization, $\mathbf{v}_{\mathrm{ref}}$ denotes the reference latent variable, $\mathcal{F}$ denotes the MT forward operator, and $\mathbf{d}^{\mathrm{obs}}$ denotes the measured MT data. Equation 12 further becomes

$$\mathcal{L}(\mathbf{v}) = \frac{1}{\|\mathbf{W}_d\mathbf{d}^{\mathrm{obs}}\|^2}\|\mathbf{W}_d(\mathbf{d}^{\mathrm{obs}} - \mathcal{F}\circ\mathcal{D}(\mathbf{v}))\|^2 + \alpha_v\|\nabla_v\mathcal{D}(\mathbf{v})\|^2 + \alpha_h\|\nabla_h\mathcal{D}(\mathbf{v})\|^2 + \lambda\|\mathbf{v} - \mathbf{v}_{\mathrm{ref}}\|^2, \tag{13}$$

where α_v, α_h, ∇_v, and ∇_h denote the regularization coefficients and vertical and horizontal gradient operators, respectively. The initial latent variable is obtained by encoding the initial model using the VAE encoder, i.e.,

$$\mathbf{v}_{\mathrm{init}} = \mathcal{E}(\mathbf{m}_{\mathrm{init}}). \tag{14}$$

To stabilize the inversion in the early stage, we use the latent variable of the $(k-1)$th model as the reference $\mathbf{v}_{k,\mathrm{ref}}$ for the kth iteration:

$$\mathbf{v}_{k,\mathrm{ref}} = \mathcal{E}\circ\mathcal{D}(\mathbf{v}_{k-1}). \tag{15}$$

Equation 13 is solved iteratively using a gradient-based method (Ellis and Oldenburg, 1994; Li and Oldenburg, 1996, 1998), specifically, the Gauss-Newton method (Habashy and Abubakar, 2004). In the kth iteration, the increment $\mathbf{p}_k$ to the current latent variable $\mathbf{v}_k$ is computed as the solution to the linear equation system:

$$\mathbf{G}(\mathbf{v}_k)\mathbf{p}_k = -\mathbf{g}(\mathbf{v}_k), \tag{16}$$

where $\mathbf{g}(.)$ and $\mathbf{G}(.)$ denote the first and second derivatives of the objective function, respectively. Adopting the chain rule, we have the following expressions:

$$\mathbf{g}(\mathbf{v}_k) = -\frac{1}{\|\mathbf{W}_d\mathbf{d}^{\mathrm{obs}}\|^2}\mathbf{J}_{\mathcal{D}}^T\mathbf{J}_{\mathcal{F}}^T(\mathbf{W}_d(\mathbf{d}^{\mathrm{obs}} - \mathcal{F}\circ\mathcal{D}(\mathbf{v}_k))) + \alpha_v\mathbf{J}_{\mathcal{D}}^T\nabla_v^T\nabla_v\mathcal{D}(\mathbf{v}_k) + \alpha_h\mathbf{J}_{\mathcal{D}}^T\nabla_h^T\nabla_h\mathcal{D}(\mathbf{v}_k) + \lambda(\mathbf{v}_k - \mathbf{v}_{k,\mathrm{ref}}), \tag{17}$$

$$\mathbf{H}(\mathbf{v}_k) = \frac{1}{\|\mathbf{W}_d\mathbf{d}^{\mathrm{obs}}\|^2}\mathbf{J}_{\mathcal{D}}^T\mathbf{J}_{\mathcal{F}}^T\mathbf{J}_{\mathcal{F}}\mathbf{J}_{\mathcal{D}} + \alpha_v\mathbf{J}_{\mathcal{D}}^T\nabla_v^T\nabla_v\mathbf{J}_{\mathcal{D}} + \alpha_h\mathbf{J}_{\mathcal{D}}^T\nabla_h^T\nabla_h\mathbf{J}_{\mathcal{D}} + \lambda\mathbf{I}, \tag{18}$$

where $\mathbf{J}_{\mathcal{F}}$ and $\mathbf{J}_{\mathcal{D}}$ are the Jacobian matrices of forward operator $\mathcal{F}$ and decoder $\mathcal{D}$, respectively. These matrices can be computed using the adjoint method (McGillivray et al., 1994) and AD in the deep-learning framework (van Merriënboer et al., 2018).

Up to now, the parametric transform between $\mathbf{m}$ and $\mathbf{v}$ has been modeled with a single VAE decoder $\mathcal{D}$ for 2D variables. With this modeling, the elements of $\mathbf{m}$ and $\mathbf{v}$ are globally linked, and the Jacobian matrix $\mathbf{J}_{\mathcal{D}}$ of the projection $\mathcal{D}$,

$$(\mathbf{J}_{\mathcal{D}})_{ij} = \frac{\partial\mathcal{D}(\mathbf{v})_i}{\partial\mathbf{v}_j}, \tag{19}$$

is a full matrix. However, this arrangement is not unique, and a generalized transform $\tilde{\mathcal{D}}$ can be introduced without loss of generality. Here, we denote the entire inversion region as DOI and the interested subregion inside the entire inversion region as ROI, as shown in Figure 1. With $\tilde{\mathcal{D}}$, the elements of $\mathbf{m}$ and $\mathbf{v}$ are locally linked through n subdomains, which are continuous yet noncoincident areas divided from the DOI. Each submodel and sublatent variable are denoted as $\mathbf{m}_i$ and $\mathbf{v}_i$, respectively, and we have

$$\mathbf{m} = (\mathbf{m}_0^T, \mathbf{m}_1^T, \mathbf{m}_2^T, \ldots, \mathbf{m}_{n-1}^T)^T, \quad \mathbf{v} = (\mathbf{v}_0^T, \mathbf{v}_1^T, \mathbf{v}_2^T, \ldots, \mathbf{v}_{n-1}^T)^T \tag{20}$$

where T denotes the transpose and $\tilde{\mathcal{D}}_i$ denotes the ith local projection between $\mathbf{m}_i$ and $\mathbf{v}_i$ with

$$\mathbf{m}_i = \tilde{\mathcal{D}}_i(\mathbf{v}_i). \tag{21}$$

The Jacobian matrix $\mathbf{J}_{\tilde{\mathcal{D}}}$ becomes

$$\mathbf{J}_{\tilde{\mathcal{D}}} = \mathrm{Diag}(\mathbf{J}_{\tilde{\mathcal{D}}_0}, \mathbf{J}_{\tilde{\mathcal{D}}_1}, \ldots, \mathbf{J}_{\tilde{\mathcal{D}}_{n-1}}). \tag{22}$$

It is a block diagonal matrix and $\mathbf{J}_{\tilde{\mathcal{D}}_i}$ denotes the Jacobian matrix of the ith local projection $\tilde{\mathcal{D}}_i$.

Suppose that a priori knowledge is spatially limited to inside ROI, which consists of subdomains $\Omega_1, \Omega_2, \ldots, \Omega_{n-1}$, whereas no a priori knowledge is available outside ROI (i.e., in the subdomain Ω_0). Therefore, the parametric transform set $\tilde{\mathcal{D}}_i$ is a combination of the VAE decoders $\tilde{\mathcal{D}}_i$ inside ROI and a direct mapping outside ROI, given by

$$\mathbf{m}_i = \mathcal{D}_i(\mathbf{v}_i), \quad i = 1, 2, \ldots n-1, \quad \mathbf{m}_i = \mathbf{v}_i, \quad i = 0. \tag{23}$$

As no a priori knowledge is available for $\mathbf{m}_0$, it is reasonable to assume that $\mathbf{m}_0$ (and also $\mathbf{v}_0$) follows a Gaussian prior. Thus, with this arrangement of $\tilde{\mathcal{D}}$, the a priori distribution of $\mathbf{v}$ remains Gaussian, and the deterministic optimization of $\mathbf{v}$ (equations 13–18) remains valid, except for the substitution of the single VAE decoder $\mathcal{D}$ with a generalized transform $\tilde{\mathcal{D}}$.

The layout of $\tilde{\mathcal{D}}$ is closely related to the spatial partitioning of the entire DOI, as well as the specific characteristics of the problem and the type of a priori knowledge being used. To improve depth resolution, we investigate a 1D subdomain encoding scheme. We divide the entire DOI into two areas: one with a priori knowledge (i.e., ROI) and one without. The former is further vertically partitioned into multiple 1D subdomains, and each subdomain, i.e., the resistivity-depth model, is encoded separately using a VAE. To reduce the complexity and take full advantage of the VAE's ability to generate multiple patterns (Lopez-Alvis et al., 2022), we compress the parametric transforms for all subdomains inside the ROI into a single VAE decoder $\mathcal{D}_0$ with

$$\mathbf{m}_i = \mathcal{D}_0(\mathbf{v}_i), \quad i = 1, 2, \ldots, n-1. \tag{24}$$

That is, one single $\mathcal{D}_0$ for 1D variables is applied multiple times to different subdomains. Similarly, the calculation of the initial latent variables (equation 14) becomes

$$\mathbf{v}_{i,\mathrm{init}} = \mathcal{E}_0(\mathbf{m}_{i,\mathrm{init}}), \quad i = 1, 2, \ldots, n-1, \quad \mathbf{v}_{i,\mathrm{init}} = \mathbf{m}_{i,\mathrm{init}}, \quad i = 0. \tag{25}$$

We denote the Jacobian matrix of the ith subdomain as $\mathbf{J}_{\mathcal{D}_0}^i$, and outside the ROI, the Jacobian matrix is the identity matrix $\mathbf{I}$. Therefore, the Jacobian matrix for the entire research area can be expressed as

$$\mathbf{J}_{\tilde{\mathcal{D}}} = \mathrm{Diag}(\mathbf{I}, \mathbf{J}_{\mathcal{D}_0}^1, \mathbf{J}_{\mathcal{D}_0}^2, \ldots, \mathbf{J}_{\mathcal{D}_0}^{n-1}). \tag{26}$$

The introduction of the proposed method in this section is summarized in Figure 3, which provides a flowchart of the method. The upper part of Figure 3 (above the dashed purple line) represents the parameterization of the pixels, whereas the bottom illustrates the inversion process. The latent variable space is constructed using VAE latent variables and the original pixels, corresponding to the pixels inside the ROI and outside the ROI, respectively. After the model parameterization is completed, the latent variables are simultaneously updated in the iterative Gaussian Newton optimization (see the loop in the lower part on the right), and the update direction for each iteration is obtained by cascading the Jacobian matrix of the forward problem and the VAE decoder. When the data residual converges, the VAE latent variables are remapped back into the pixel space, and a final model in the pixel space is constructed.

NUMERICAL TESTS

Experiments with strong a priori knowledge

In this section, the effectiveness of the proposed method is verified with synthetic experiments. First, we assume that the a priori

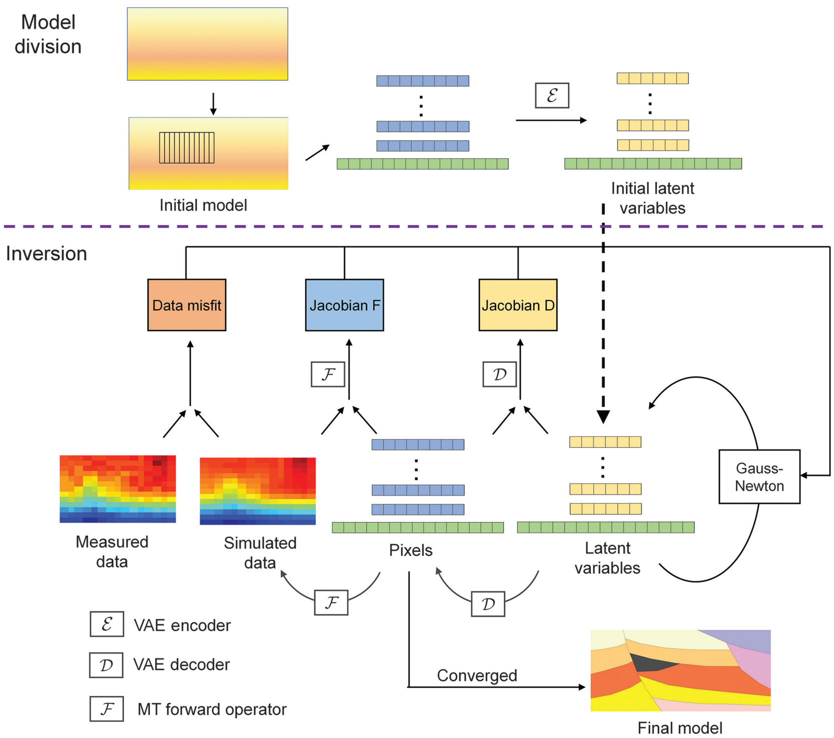

Figure 3. The diagram of the proposed feature-based inversion.

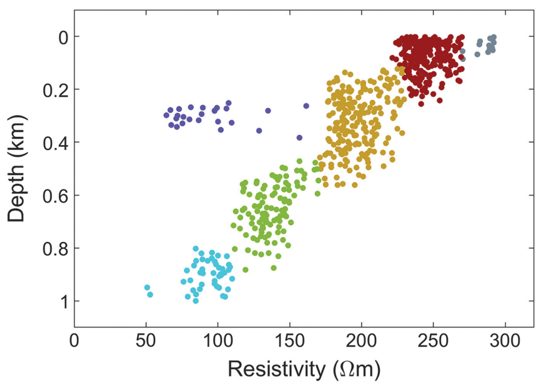

Figure 4. The resistivity-depth distribution derived from well logs.

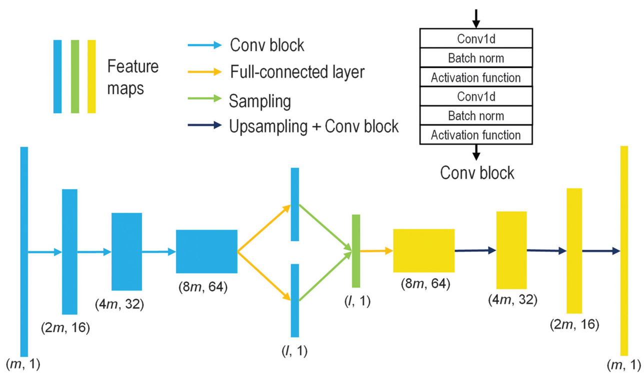

Figure 5. The structure of the VAE adopted in this work. The feature maps of the encoder and decoder are colored blue and yellow, respectively.

knowledge is available in the entire DOI with low uncertainty. We also assume the availability of well-logging data in the inversion area, providing information on the subsurface resistivity-depth distribution (as shown in Figure 4). The distribution can be divided into six clusters, with data density describing the frequency of the sampled attribute. Based on the assumption of a layered structure, we infer that the subsurface formation consists of four layers, with resistivity mainly decreasing with depth. The layer boundaries are located at depths of approximately 250, 550, and 850 m. There are two attribute clusters with lower frequency, denoted with gray and purple, indicating the existence of two small abnormal blocks located at approximately 300 m deep and near the surface, respectively.

According to Figure 4, we designed the data set as follows: the three-layer boundaries vary from 100 to 400 m, 450 to 650 m, and 750 to 950 m, respectively. The resistivity (Ωm) of the four layers is sampled from uniform distributions $\mathcal{U}(230, 270)$, $\mathcal{U}(180, 220)$, $\mathcal{U}(110, 150)$, and $\mathcal{U}(60, 100)$. We also set a near-surface high-resistivity layer below 130 m and a conductive anomaly at approximately 300 m deep with resistivity sampled from $\mathcal{U}(280, 320)$ and $\mathcal{U}(110, 150)$, respectively, with a probability of 0.2. We generated 30,000 1D training models, each of which was blurred with a 1D Gaussian filter to smooth the sharp model boundary and improve the neural network convergence. The length of each model is 32, and the input size and latent dimension of the VAE are 32 and 16, respectively (see Figure 5). We trained the VAE on a 48 kernel 2.3 GHz Linux server with 512 G RAM and 4 NVIDIA V100 GPUs, and the total training time was approximately 1 h. We show 50 cases sampled from the data set and the convergence curve of the VAE network in Figure 6. For clarity, we have referred to this VAE network as Net-A in the following.

Figure 7a–7c shows three ground truth models for inversion, referred to as model A, model B, and model C, respectively. These models reflect similar a priori knowledge but with significant variations in the trend of the layer boundaries and the number and shape of the anomaly blocks. In this work, we used the 2D code of transverse electric (TE) mode for MT forward modeling in all synthetic and field experiments. We used 14 frequencies ranging from 3.2 Hz to 1×10^4 Hz and 16 receivers uniformly distributed on the surface. The synthetic apparent resistivity and impedance phase were calculated and corrupted with 5% Gaussian noise, as shown in Figure 7d–7i. We constructed the initial latent variable by replicating the latent variable of a 100 Ωm uniform 1D depth model. The inverted models of the pixel-based approach, feature-based approach, and convergence curves are shown in Figure 8 and the reconstructed data are shown in Figure 9.

The pixel-based inversion recovers the inverse relationship between resistivity and depth and accurately reconstructs the location of the two anomalies. However, the shape and attribute value of the central conductor cannot be clearly reconstructed, and the boundaries between different layers are difficult to recognize, with unclear boundary trends.

In the feature-based inversion, the trends of the layers are reconstructed clearly, and the shapes of the anomalies are recovered better. The VAE generates appropriate models whose patterns conform to a priori information, and the gradient-descent method selects a model that simultaneously satisfies the a priori information and minimizes data misfit. The VAE reduces the model space reasonably and facilitates the search for the appropriate model. Therefore, the inversion takes fewer steps to converge with better data fitting. The feature-based inversion converges in approximately four steps, slightly less than the pixel-based inversion. The final data misfit E_d and model misfit E_m are calculated by

$$E_d = \frac{\|\mathbf{d}_{\text{inv}} - \mathbf{d}_{\text{tru}}\|_2}{\|\mathbf{d}_{\text{tru}}\|_2}, \quad E_m = \frac{\|\mathbf{m}_{\text{inv}} - \mathbf{m}_{\text{tru}}\|_2}{\|\mathbf{m}_{\text{tru}}\|_2}, \tag{27}$$

where $\mathbf{d}_{\text{inv}}$, $\mathbf{d}_{\text{tru}}$, $\mathbf{m}_{\text{inv}}$, and $\mathbf{m}_{\text{tru}}$ denote the inverted data, ground truth data, inverted model, and ground truth model, respectively, and $\|\cdot\|_2$ denotes the L_2-norm. The final data and model misfits for the inversion of models A, B, and C are shown in Table 1. The two approaches lead to similar data misfits ultimately, but the model misfit of the feature-based inversion is merely a quarter of the pixel-based approach.

Experiments with more uncertain a priori knowledge

To further verify the proposed algorithm, we consider scenarios in which the a priori knowledge is less informative. We assume that the a priori knowledge in the previous experiment is still valid, except for some corruption in three different aspects.

1) The attribute value of two anomaly blocks is more uncertain.
2) The depth of the conductive block is more uncertain.
3) The layer number and the existence of abnormal blocks are unknown.

The data set generation details have been revised based on the corrupted a priori knowledge. In the first scenario, the resistivity (Ωm) of the near surface and central blocks are sampled from a uniform distribution $\mathcal{U}(20, 320)$. In the second scenario, the depth (m) of the conductive block is sampled from a uniform distribution $\mathcal{U}(130, 550)$ and the thickness (m) from $\mathcal{U}(100, 200)$. In the third

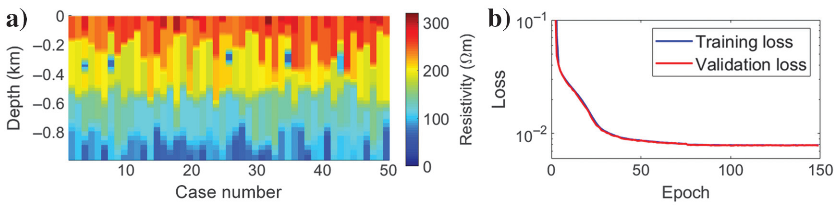

Figure 6. (a) The 50 cases of the data set for Net-A and (b) the convergence of training and validation loss.

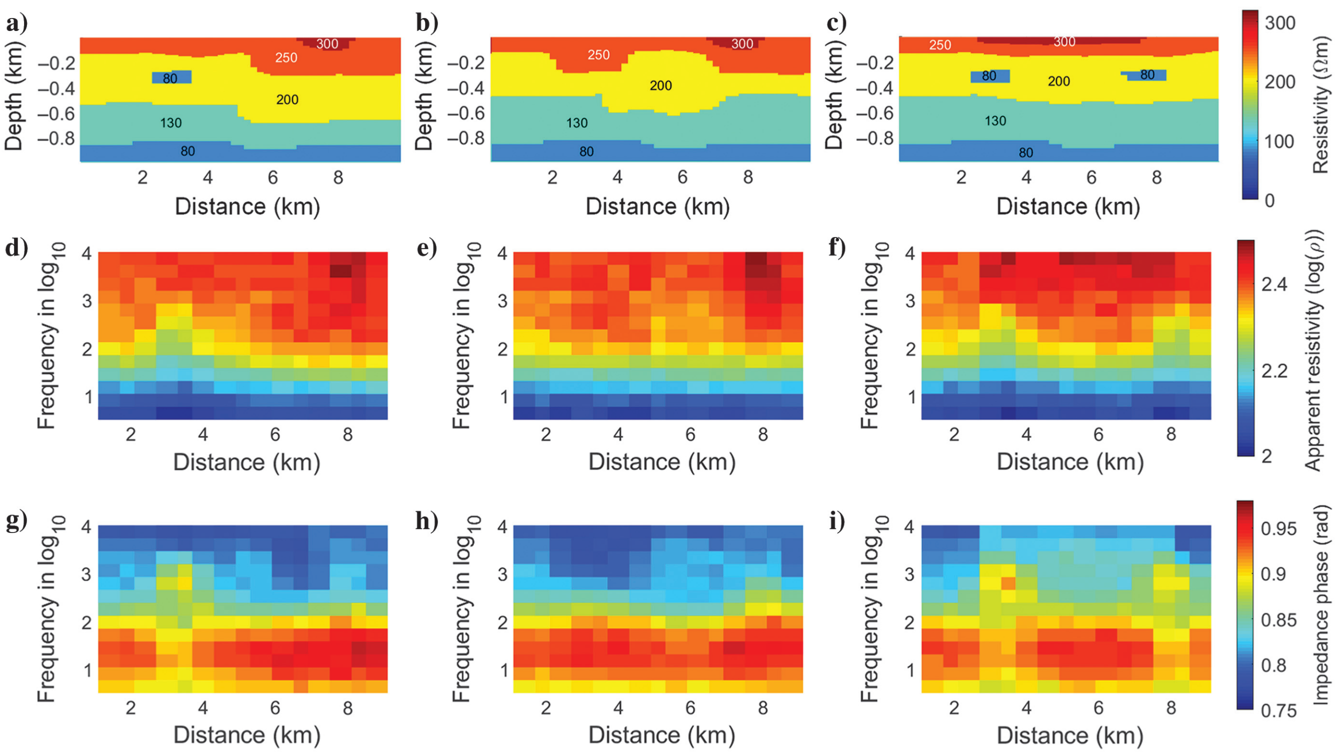

Figure 7. (a, d, and g), (b, e, and h), and (c, f, and i) Ground truth models, corresponding apparent resistivity, and impedance phase for models A, B, and C, respectively.

scenario, the layer number varies between 10 and 15, with the resistivity (Ωm) varying from 300 to 30 with depth. Thirty percent of the cases have an anomaly that may reside at any depth, and the resistivity (Ωm) is also sampled from $\mathcal{U}(30, 300)$. We generate 30,000 training models for each scenario, all smoothed with a Gaussian filter. Three VAEs, referred to as Net-B, Net-C, and Net-D, respectively, are trained using three data sets. The structure of the VAEs used here is identical to that used previously, and the total training time for each scenario is approximately 1 h. Figure 10 shows some examples of the data sets and the training loss curves.

The feature-based approach is used to invert model A with the Net-B, C, and D VAEs trained from the updated data sets, and the results are shown in Figure 11. Despite the weaker a priori knowledge, the models reconstructed with Net-B and Net-C are similar to that with Net-A. The shapes and positions of the two anomaly blocks are accurately depicted, and the trend of layer boundaries is characterized by high resolution. With the weakening of a priori information, the VAE yields a more complex and diverse model space. However, the low sensitivity of MT data hampers the optimization method in searching the global minima from an expanded model space, leading to a conductive artifact in the third layer at an offset of approximately 2.5 km. With Net-D, the shallower formations, such as the resistive anomaly and the boundary between the first and second layers, can still be recovered with relatively high accuracy and resolution. However, the deeper formations, such as the layer boundaries between the second, third, and fourth layers, and the conductive block have a blurry depiction with limited resolution. The higher data sensitivity of shallower formations guarantees that the global minimum can be searched from an enlarged model space. Nevertheless, such a model space fails to provide sufficiently informative a priori knowledge for the reconstruction of deep formations because data sensitivity is lower, yielding a smoothed-out deep structure similar to the pixel-based approach.

Table 2 shows the final data and model misfits for the pixel- and feature-based inversion using four VAE networks. The convergence curves of the four feature-based approaches are shown in Figure 12, and all of them converge in approximately five steps. The final data misfits are 0.0056, 0.0058, 0.0054, and 0.0054, respectively, and the final model misfits are 0.023, 0.022, 0.025, and 0.028, respectively. These misfit values are all significantly smaller than that obtained by the pixel-based inversion (0.10).

Experiments using a priori knowledge with limited spatial scope

The available a priori knowledge in the following experiments is limited to a subregion (ROI). We either have no prior understanding of the geophysical structure outside ROI, or the a priori understanding is too uncertain to provide informative constraints. Suppose the rectangular subregion ROI has a depth ranging from 200 to 400 m. A local conductive anomaly may exist at any depth with a probability of 0.3. The background resistivity ranges from 180 to 220 Ωm, and the anomaly resistivity varies from 60 to 100 Ωm, both obeying the uniform distribution. We generate 30,000 training models, each smoothed with a Gaussian filter same as previously. The VAE structure is identical to that used previously, except that the input dimension is 16 and the latent dimension is eight. The total training time is approximately 30 min. Figure 13 shows 50 cases of

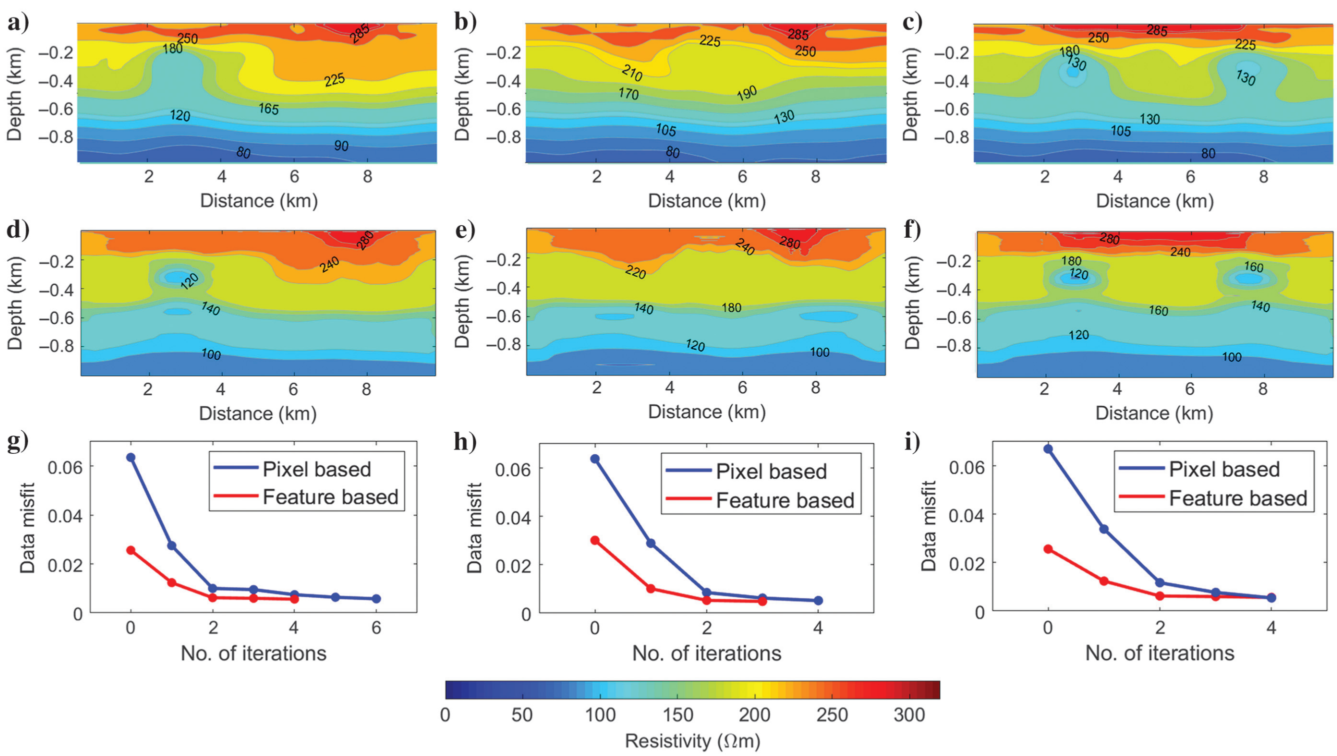

Figure 8. (a and d), (b and e), and (c and f) The pixel- and feature-based reconstructions of models A, B, and C, respectively. (g–i) The convergence curves using the two approaches of models A, B, and C, respectively.

the data set and the loss convergence curve. For clarity, we refer to this VAE as Net-E.

The two ground truth models are shown in Figure 14, and the rectangular box represents the ROI. The first model is the previously defined model C, and the second model (referred to as model D) is the same as model C inside the rectangular ROI but differs outside the ROI. The pixel- and feature-based approach with Net-A and E are used to reconstruct both models. In the feature-based approach with Net-A, the models are parameterized by the latent variables of the entire depth range, whereas in the feature-based approach with Net-E, the models are parameterized by the latent variables inside and the pixels outside the rectangular box. The reconstructions of the pixel- and feature-based inversion with Net-A and Net-E are shown in Figure 15. The pixel-based approach

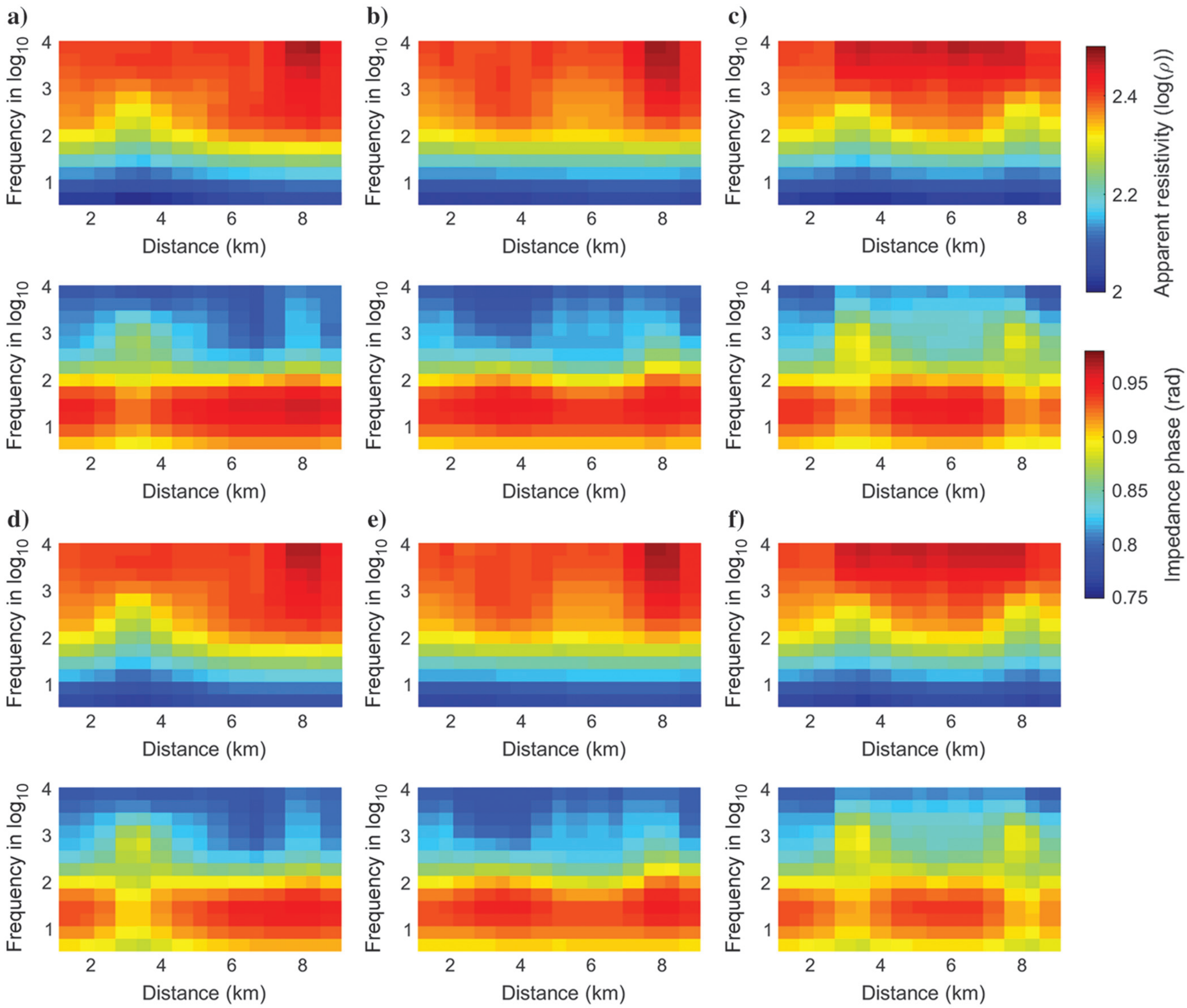

Figure 9. (a–c) The inverted data with the pixel-based approach and (d–f) the inverted data with the feature-based approach.

Table 1. Evaluation of inverted models A, B, and C using Net-A.

Model	E_d (pixel based)	E_m (pixel based)	E_d (feature based)	E_m (feature based)
Model A	0.0057	0.10	0.0056	0.023
Model B	0.0051	0.098	0.0048	0.019
Model C	0.0054	0.10	0.0054	0.024

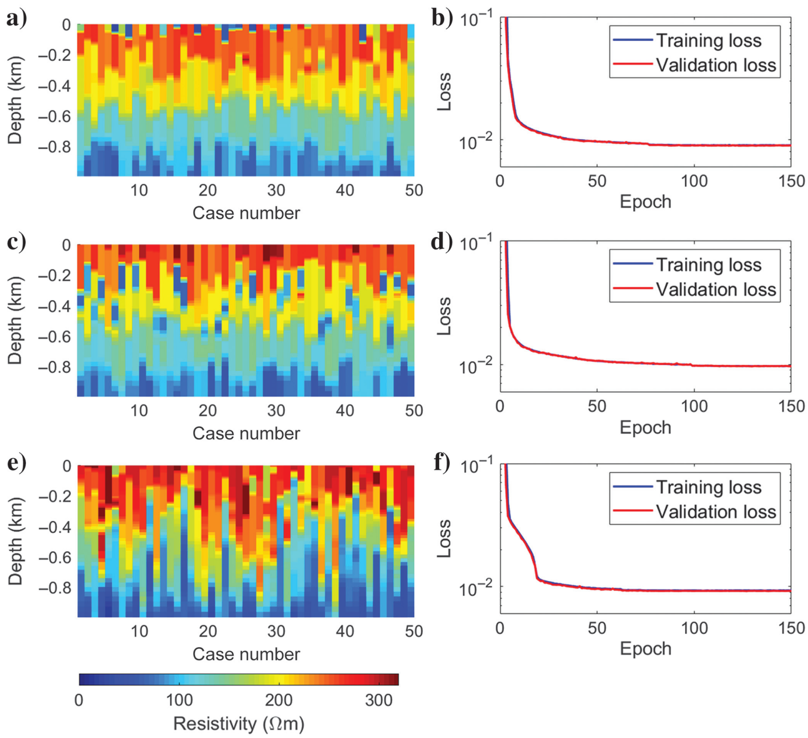

Figure 10. (a, c, and e) The 50 cases of three data sets with weakened knowledge and (b, d, and f) corresponding convergence of training and validation loss.

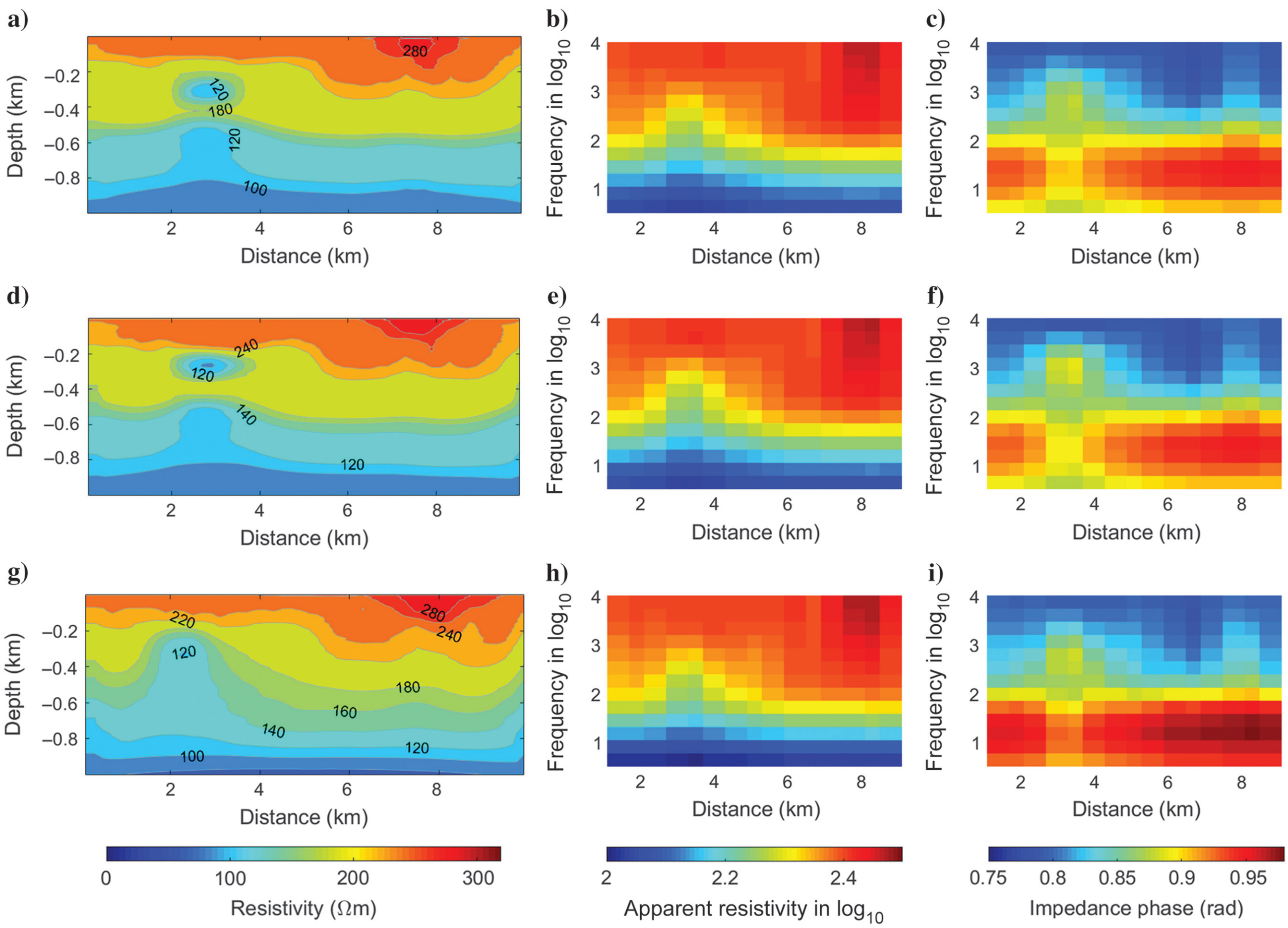

Figure 11. (a–c) The inverted apparent resistivity and impedance phase with Net-B and those with (d–f) Net-C and (g–i) Net-D, respectively.

can correctly reconstruct the trend of resistivity variation, but the details of the underground structures are unclear. The feature-based inversion with Net-A yields an accurate reconstruction of model C, with all subsurface formations resembling the ground truth. However, the feature-based reconstruction with Net-A shown in Figure 15d erroneously reconstructs a resistive block at approximately 800 m deep as a conductive layer. The Net-A constrains the reconstruction of model D with the knowledge learned from the training data set, which is incompatible with the real case.

With the feature-based inversion using Net-E, the available a priori knowledge enhances the characterization of conductive blocks only within the encoded subdomain. This results in more accurate boundary and attribute values for conductive blocks in models C and D compared with the pixel- and feature-based inversion using Net-A. However, the reconstruction of formations outside the ROI (e.g., below 400 m) resembles that of the pixel-based approach, with lower resolution and resistivity distribution due to a lack of a priori knowledge. Despite this, the subdomain encoding scheme helps avoid erroneous local minima when global knowledge is improper.

The data misfit convergence curves of the two models are shown in Figure 15g and 15h. For model C, the final data misfits of the three inversions are similar, all approximately 0.0055. The final model misfit of the feature-based approach using Net-A and Net-E is also similar and significantly lower than the pixel-based method. For model D, the final data misfit of the feature-based method using Net-A is approximately 0.02, significantly larger than that of the pixel-based and feature-based methods using Net-E. The model misfit of the feature-based method using Net-E is the least among the three reconstructions. The final data and model misfits of the two models are provided in Table 3.

Table 2. Evaluation of inverted model A with Net-A, B, C, and D.

Network	E_d	E_m
Net-A	0.0056	0.023
Net-B	0.0058	0.022
Net-C	0.0054	0.025
Net-D	0.0054	0.028

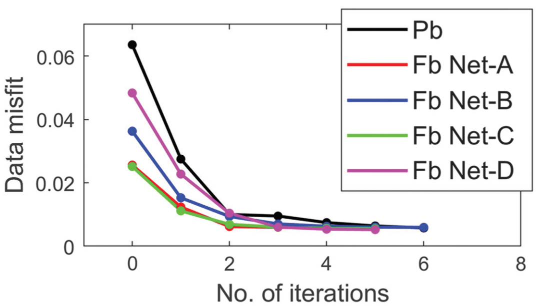

Figure 12. The convergence curves of the pixel-based approach and feature-based approaches with Net-A, B, C, and D. Here, Pb and Fb in the legend are abbreviations of pixel based and feature based, respectively.

APPLICATION TO FIELD DATA

We apply the proposed feature-based inversion algorithm to field data. The field data are collected by the SAMTEX project (Jones et al., 2009), a multinational scientific project that applies MT to understand the tectonic process in southern Africa. The project includes more than 750 MT sites and covers the main tectonic structures across southern Africa. Among them, the Etosha-to-Kimberly (ETO-KIM) profile is located in the northwest of the survey area, crossing three main geologic structures, namely Congo Craton, Damara-Ghanzi-Chobe Belt, and Rehoboth Craton. This profile is approximately 760 km in length, oriented in the northwest–southeast direction, and located near the west coast of Africa (see Figure 16). The earlier study (Khoza et al., 2013) inverts the ETO-KIM profile with isotropic and anisotropic 2D MT inversion and 3D inversion and provides an insight into the lithospheric structure of the Archean craton. Following the Khoza et al. (2013), we conduct a data inversion of the ETO-KIM profile with the proposed feature-based approach.

Thirty-six receivers were deployed along the ETO-KIM profile, with 26 located in the northern ETO section and 10 in the southern KIM section. The MT data collected from the KIM profile consisted of 80 frequencies ranging from 3.4×10^{-4} Hz to 320 Hz. In addition, four lower frequencies were collected at ETO receivers.

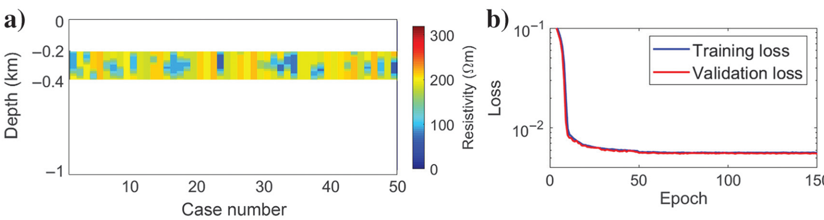

Figure 13. (a) Some cases of the data set for Net-E and (b) the convergence of training and validation loss.

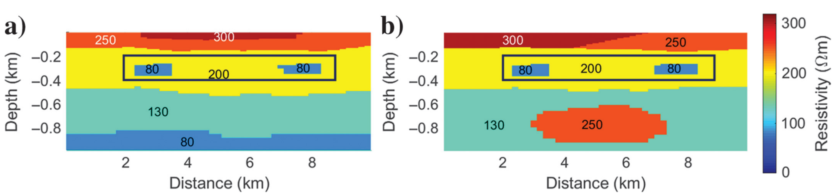

Figure 14. (a) Model C and (b) model D with the same rectangular encoding regions.

However, the measured pseudosection at low frequency had poor quality with some bad points and a significant static shift effect. Therefore, only 20 frequencies ranging from 5×10^{-3} Hz to 320 Hz were used for the inversion. The bad points were removed and the static shift correction was conducted using the median filtering method (Sternberg et al., 1982). The MT pseudosections used in the inversion are shown in Figure 17, with the pixels corresponding to bad points left blank.

The MT forward problem is solved using a mesh with 64 pixels in depth and 100 pixels in the offset direction, with layer thickness increasing in a ratio of 1.146. The thickness of the most shallow and deepest layers is 2 m and approximately 10 km, respectively. This mesh configuration ensures a small discrepancy between MT data calculated by the numerical algorithm and theoretical derivation in the 1D case. The pixel-based reconstruction is shown in Figure 18a. The inversion converges in seven steps, with a final data misfit of 0.097. The reconstructed apparent resistivity and impedance phase are shown in Figure 19c and 19d, respectively. The mid-upper resistive layer and the mid-lower crustal conductors in the central zone, ranging from offset 90 to 430 km, and the high-resistivity block in the northern zone are the major geophysical features observed in the inversion. These features correspond to Pan-African granites, the crustal conductors, and the Archean basement called Kamanjab Inlier (Khoza et al., 2013). However, due to the limited penetrative ability of the EM fields below the conductive body, the lower boundary of the mid-lower crustal conductors cannot be well resolved, which is observed in the work of Khoza et al. (2013) and our pixel-based inversion.

We aim to use the feature-based method to improve the delineation of the lower boundary of the mid-lower crustal conductors. However, we have not found any existing research with a priori knowledge about the geologic stratification of the survey area. Therefore, we introduce a hypothetical a priori knowledge that the average thickness of the mid-lower conductor is approximately

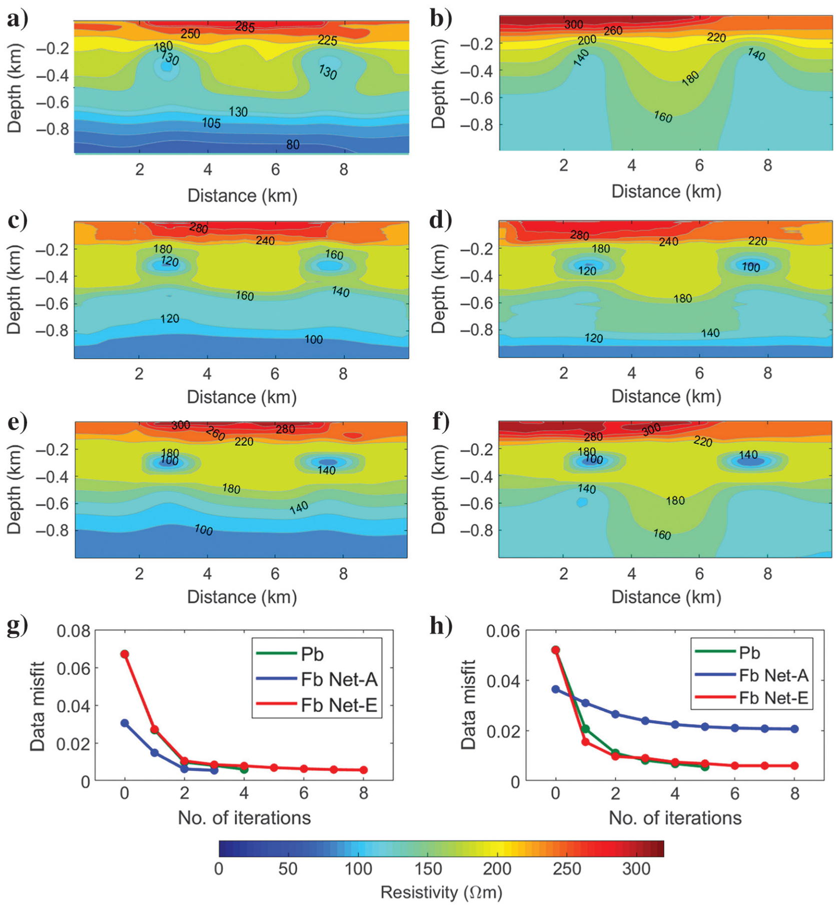

Figure 15. The inversion of model C and model D. (a and b) the pixel-based inversion result, (c and d) the feature-based reconstruction with Net-A, (e and f) the feature-based reconstruction with Net-E, and (g and h) the convergence curves. Here, Pb and Fb in the legend are abbreviations of pixel based and feature based, respectively.

Table 3. Evaluation of inverted models C and D with pixel-based inversion and feature-based inversion with Net-A and Net-E.

Model	E_d (pb)	E_m (pb)	E_d (fbA)	E_m (fbA)	E_d (fbE)	E_m (fbE)
Model C	0.0054	0.10	0.0054	0.024	0.0056	0.024
Model D	0.0055	0.094	0.021	0.040	0.0059	0.031

Note: pb, fbA, and fbE are abbreviations of pixel-based inversion, feature-based inversion with Net-A, and feature-based inversion with Net-E, respectively.

15 km. The ROI covers 40 pixels horizontally and 32 pixels vertically, ranging from 90 to 430 km in an offset direction and 1–80 km in depth. A schematic diagram of the ROI and the area outside it is shown in Figure 20, with the blue rectangle representing the encoded ROI and the red wavy lines indicating the area outside the ROI.

To generate the training data set, we sample the depth of the top edge and the thickness of the mid-lower conductors (in km) from a uniform distribution $\mathcal{U}(2, 20)$, and the logarithmic resistivity (Ωm) from $\mathcal{U}(-1.5, 1.5)$. Below the lower boundary, the logarithmic resistivity ranges from 2.6 to 4.5, increasing with depth. We generate a total of 60,000 1D training models. The VAE structure used is the same as previously, except that the input dimension is 32 and the latent dimension is 16. The VAE is trained for approximately 1 h. Figure 21a shows some examples of the training set, whereas Figure 21b shows the training and validation loss of the VAE.

We perform a feature-based inversion of the ETO-KIM profile data, and as shown in Figure 18b, a more distinct lower boundary of the mid-lower conductor is depicted at approximately 40 km depth, compared with the pixel-based inversion, whereas the reconstruction outside the ROI remains largely unchanged. The feature-based inversion converges in nine steps, with a final data misfit of 0.096. The inversion convergence curves are shown in Figure 22. The proposed algorithm incorporates a priori knowledge about the thickness of mid-lower crustal conductors into the inversion. The inverted model is consistent with the a priori knowledge, with minimal changes in the data fitting. It is important to note that the a priori knowledge used in this study is hypothetical, and the reconstructed model may not accurately reflect the true geologic structure. In future field applications with reliable knowledge, the proposed algorithm may provide reconstructions that better match the actual geologic structure.

DISCUSSION

One advantage of feature-based inversion is the flexibility of embedding a priori information. First, the prior distribution may present a rather complex manifold due to the possibly complex subsurface formations and attribute distribution. The VAE is capable of regularizing a complex distribution to Gaussian distribution and generating new models well consistent with the a priori knowledge. Second, the inversion performance can be improved provided that the proper model pattern (rather than the accurate model) is included in the model space. In the feature-based inversion, the reconstruction accuracy depends on the coaction of data sensitivity and the a priori knowledge embedded in the VAE. With a certain level of data sensitivity, the reconstructed model will become more accurate as the a priori information is strengthened (see Figures 8d, 11a, 11d, and 11g). On the other hand, based on data sensitivity, to different degrees the different parts can be enhanced by the embedded a priori information (e.g., the resolution varying with depth in Figure 11g). It is worth noting that in the feature-based inversion, instead of being useless or inevitably leading to implausible local minima, the weak a priori knowledge brings a large a priori feasible domain, and from which searching the global minimum is still possible. Hence, the feature-

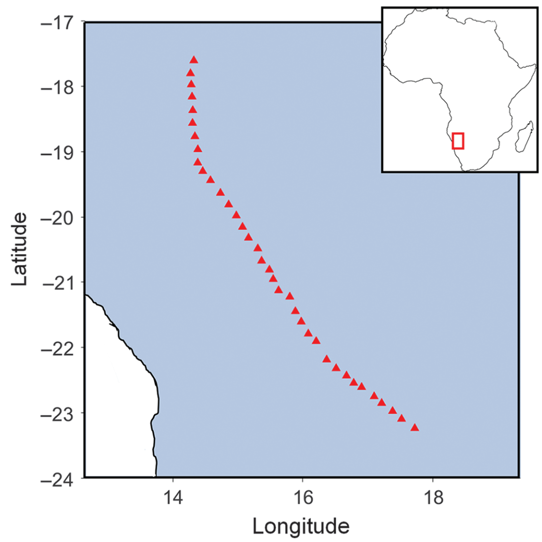

Figure 16. The receivers (the red triangles) used in this study. The red box in the inset map denotes the location of the research area (Jones et al., 2009).

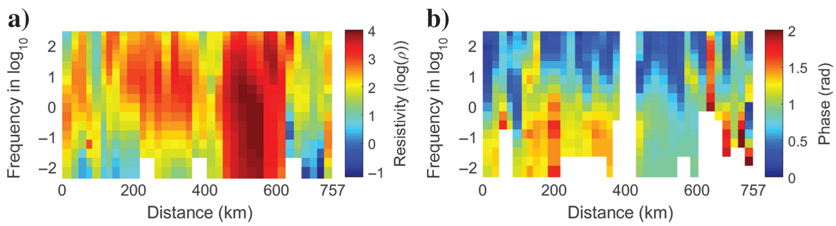

Figure 17. (a and b) The processed TE-mode apparent resistivity and impedance phase.

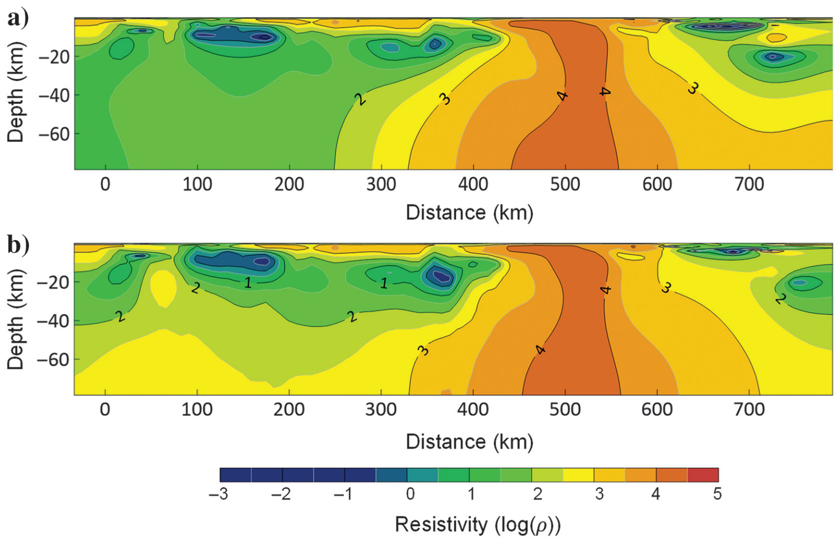

Figure 18. (a) The pixel-based and (b) feature-based reconstruction of the ETO-KIM profile.

based inversion possesses strong stability and high self-adaptation ability. Third, the encoded ROI can be flexibly determined according to the problem characteristics. The attribute distribution and resolution resemble that of pixel-based inversion outside the ROI, whereas the inversion accuracy and resolution are enhanced inside the ROI. Therefore, feature-based inversion may be a possible solution for multiresolution and multiscale inversion.

Another advantage of feature-based inversion lies in the flexibility of the encoding scheme. In feature-based inversion, the ROI may consist of several disjoint subdomains with varying a priori information. One approach could be to design multiple training sets with different a priori information, train VAEs with different parameters, and use different parameterizations for each subdomain. However, this method requires training multiple VAEs, which can make the inversion process more complex. Because the VAE has a strong capability to learn complex distributions, it can generate models with very different patterns using a single VAE. By using the subdomain encoding scheme, it is possible to train a single VAE with a training data set that includes patterns from different subdomains and use this VAE to parameterize all subdomains. This approach is advantageous because the generation of training data sets becomes less challenging because only local patterns need to be included instead of global patterns, thereby reducing the diversity and complexity of the data set. In addition, the computational cost of training the neural network can be reduced. Even when the ROI is continuous with uniform a priori knowledge, dividing the DOI into various subdomains is still beneficial. In this work, we study a particular case of subdomain encoding, namely encoding the 1D resistivity-depth models separately in the 2D resistivity model. This approach is particularly useful in EM surveys where increasing the receiver density can improve horizontal resolution, but the penetration ability of EM waves limits vertical resolution and accuracy (Bedrosian, 2007). By constraining the plausible patterns of the resistivity-depth model, the 1D subdomain encoding scheme enhances the depth resolution of the MT inversion. Importantly, this encoding scheme decouples the parameterization of forward modeling and the model reparameterization scheme, allowing for the flexible extension to inversion problems of different dimensions and scales, such as other 2D or 3D problems. Moreover, the proposed method can be applied to other geophysical EM inverse problems, such as CSEM and transient EM inversion, and has potential applications in seismic explorations such as thin reservoir identification and evaluation.

We would like to highlight some important considerations regarding feature-based inversion. First, it is essential to ensure that the training data set includes the correct model pattern to obtain plausible inversion results. Adequate a priori knowledge about the survey area is required; otherwise, the feature-based inversion may not be able to outperform the pixel-based approach. Second is the generalization ability. Unlike more versatile inverse problem solvers (Guo et al.,

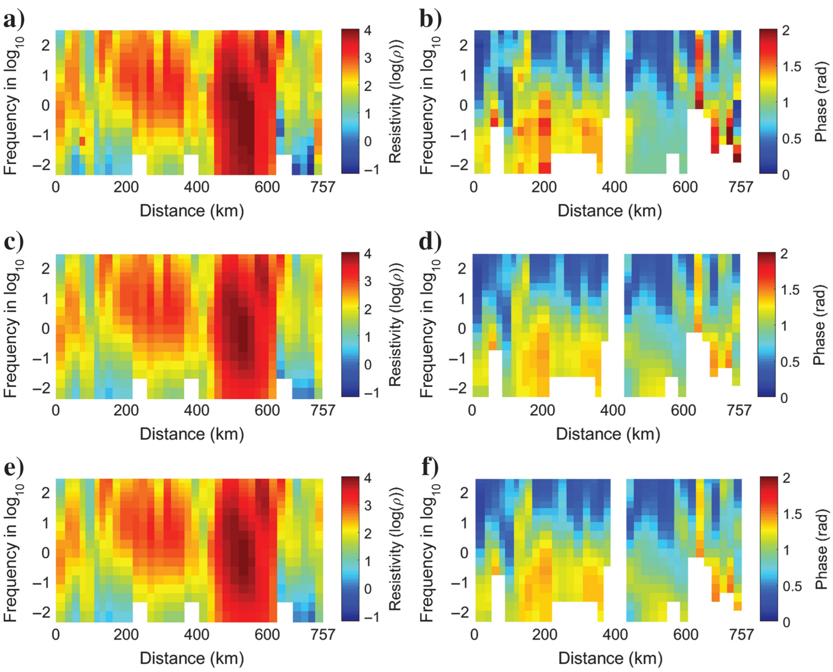

Figure 19. (a and b) Processed apparent resistivity and impedance phase, (c and d) inverted apparent resistivity and impedance phase with the pixel-based approach, and (e and f) inverted apparent resistivity and impedance phase with the feature-based approach.

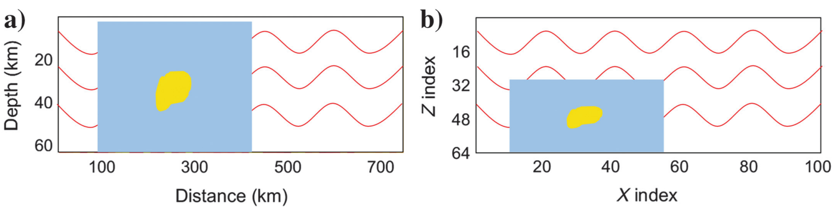

Figure 20. The schematic illustration of (a) encoding region inside the entire study area and (b) encoded pixels inside the entire pixel matrix.

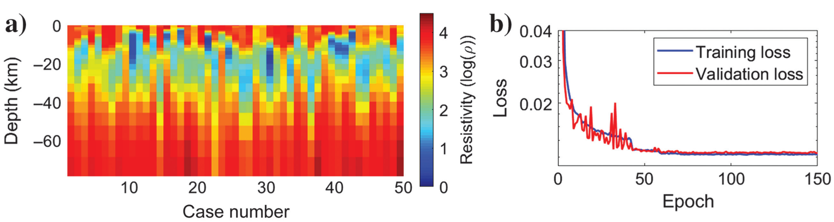

Figure 21. (a) Some cases of the data set for field application and (b) the convergence of training and validation loss.

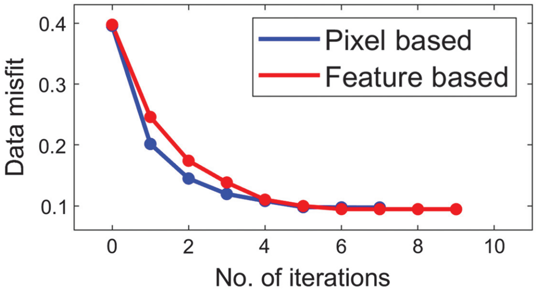

Figure 22. The convergence curve of the pixel- and feature-based approach.

2020; Sun and Alkhalifah, 2020) and problem-specific neural solvers based on physics-informed neural networks (bin Waheed et al., 2021), the feature-based inversion often requires that the real patterns are included in the a priori distribution of the training data set. Increasing the diversity of the training data set may enlarge the model space and improve the generalization ability, but this could also dilute the a priori knowledge. Third, the data sensitivities of latent variables and pixels may not match. Therefore, a simultaneous inversion of the latent variables and pixels may result in different convergence speeds and possible nonconvergence issues. One potential solution is to invert the latent variables and pixels alternatively or separately, with necessary modifications to the optimization algorithm.

We acknowledge that there are still some aspects of this study that could be further explored. First, the quantification of a priori information has not been rigorously defined. It may be possible to define the quantity of information that the VAE learns from the training data sets based on a probabilistic framework and investigate how different VAE networks can reduce inversion uncertainty. Second, due to the lack of additional field data, we cannot obtain a priori knowledge to support the feature-based inversion. We have introduced some hypothetical knowledge about the thickness of the mid-lower crustal conductor and verified the effectiveness of our algorithm, but our reconstruction may be inconsistent with reality. In the future, the algorithm can be verified with more suitable field examples to further test its effectiveness.

CONCLUSION

In this study, we propose a feature-based 2D MT data inversion method using a VAE with a subdomain encoding scheme. Instead of encoding the entire DOI, we adopt a 1D subdomain encoding scheme to encode the 1D resistivity-depth models using a single VAE. The unknowns, including the latent variables of the ROI and pixels outside the ROI, are simultaneously inverted with a gradient-descent algorithm. The ROI recovered from the latent variables exhibits a priori patterns, whereas the reconstruction outside the ROI is similar to the pixel-based approach. The 1D subdomain encoding scheme reduces the requirements for data set complexity and diversity, resulting in a relatively low training cost. The scope of the ROI can be flexibly determined according to a priori knowledge or practical demands. By reasonably designing the data set, knowledge with different uncertainties can be adaptively incorporated into the inversion. We apply the proposed method to invert synthetic data and the SAMTEX field data. Both results demonstrate that the proposed method improves inversion accuracy and resolution.

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DATA AND MATERIALS AVAILABILITY

Data associated with this research are available and can be obtained by contacting the corresponding author.

APPENDIX A

DETERMINISTIC APPROACH WITH PARAMETRIC TRANSFORM

In the stochastic inversion framework, the inverse problem is solved by maximizing the posterior distribution of model $\mathbf{m}$ given the measurement $\mathbf{d}$. The Bayesian formula reveals the substantial relationship between the prior distribution $\rho_m(\mathbf{m})$, likelihood $p(\mathbf{d}|\mathbf{m})$, and posterior probability $\sigma_m(\mathbf{m}|\mathbf{d})$ with

$$\sigma_m(\mathbf{m}|\mathbf{d}) = k \cdot p(\mathbf{d}|\mathbf{m})\rho_m(\mathbf{m}), \tag{A-1}$$

where k is a normalization constant. The likelihood $p(\mathbf{d}|\mathbf{m})$ relates to measurement uncertainties, which is a collective effect of measurement noise, environmental noise, positional errors of instruments and human factors, etc. We assume the measurement uncertainty conforms to Gaussian distribution:

$$p(\mathbf{d}|\mathbf{m}) \propto \exp\left\{-\frac{1}{2}(\mathbf{d}-\mathcal{F}(\mathbf{m}))^T\mathbf{C}_d^{-1}(\mathbf{d}-\mathcal{F}(\mathbf{m}))\right\}, \tag{A-2}$$

where $\mathcal{F}$ denotes the forward-modeling operator and $\mathbf{C}_d$ denotes the covariance matrix for the measured data. However, many realistic assumptions of the model prior $\rho_m(\mathbf{m})$ are not exponential (e.g., Gaussian or generalized Gaussian) (Tarantola, 2005). As a result, we do not assume the mathematical form of $\rho_m(\mathbf{m})$ (and thus $\sigma_m(\mathbf{m}|\mathbf{d})$) here for generality. Consequently, the deterministic inversion as introduced in Habashy and Abubakar (2004) may not be applicable.

With the aid of proper reparameterization methods, it is possible to project the model parameter $\mathbf{m}$ obeying an arbitrary distribution to some transformed parameter space obeying Gaussian distribution, for example, the latent variable space $\{\mathbf{v}\}$ of VAEs, which conforms to standard normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$ (see equation 5). Following equation A-1, we similarly have

$$\sigma_v(\mathbf{v}|\mathbf{d}) = k \cdot p(\mathbf{d}|\mathbf{v})\rho_v(\mathbf{v}), \tag{A-3}$$

where k is a normalization constant and $\rho_v(\mathbf{v})$, $p(\mathbf{d}|\mathbf{v})$, and $\sigma_v(\mathbf{v}|\mathbf{d})$ are the prior probability, likelihood, and posterior probability of the latent variable $\mathbf{v}$, respectively. Adopting the total probability formula and conditional probability formula, we have

$$p(\mathbf{d}|\mathbf{v}) = \int_{\mathbf{m}} p(\mathbf{d},\mathbf{m}|\mathbf{v})d\mathbf{m} = \int_{\mathbf{m}} p(\mathbf{d}|\mathbf{m},\mathbf{v})p(\mathbf{m}|\mathbf{v})d\mathbf{m}, \tag{A-4}$$

where $p(\mathbf{m}|\mathbf{v})$ denotes the conditional probability density of $\mathbf{m}$ given $\mathbf{v}$ and $p(\mathbf{d}|\mathbf{m},\mathbf{v})$ denotes the conditional probability density of $\mathbf{d}$ given $\mathbf{m}$ and $\mathbf{v}$. Given model $\mathbf{m}$, data $\mathbf{d}$, and latent variables $\mathbf{v}$ are conditionally independent, thus equation A-4 further becomes

$$p(\mathbf{d}|\mathbf{v}) = \int_{\mathbf{m}} p(\mathbf{d}|\mathbf{m})p(\mathbf{m}|\mathbf{v})d\mathbf{m}. \tag{A-5}$$

Here, we model the conditional probability density $p(\mathbf{m}|\mathbf{v})$ with the VAE decoder $p_\theta(\mathbf{m}|\mathbf{v})$ (see equation 7). To simplify the calculation, we assume the covariance coefficient c of $p_\theta(\mathbf{m}|\mathbf{v})$ approximates zero. Hence, $p_\theta(\mathbf{m}|\mathbf{v})$ becomes a Dirac distribution:

$$p_\theta(\mathbf{m}|\mathbf{v}) = \delta(\mathbf{m} - \mathcal{D}(\mathbf{v})). \tag{A-6}$$

Equation A-5 further becomes

$$\begin{aligned} p(\mathbf{d}|\mathbf{v}) &= \int_{\mathbf{m}} \exp\left\{-\frac{1}{2}(\mathbf{d}-\mathcal{F}(\mathbf{m}))^T\mathbf{C}_d^{-1}(\mathbf{d}-\mathcal{F}(\mathbf{m}))\right\}\delta(\mathbf{m}-\mathcal{D}(\mathbf{v}))d\mathbf{m} \\ &= \exp\left\{-\frac{1}{2}(\mathbf{d}-\mathcal{F}(\mathcal{D}(\mathbf{v})))^T\mathbf{C}_d^{-1}(\mathbf{d}-\mathcal{F}(\mathcal{D}(\mathbf{v})))\right\}. \end{aligned} \tag{A-7}$$

Substituting equation A-3 with equations 5 and A-7, the posterior distribution of $\mathbf{v}$ can be derived with

$$\begin{aligned} \sigma_v(\mathbf{v}|\mathbf{d}) = \exp\Bigg\{&-\frac{1}{2}\mathbf{v}^T\mathbf{v} \\ &-\frac{1}{2}(\mathbf{d}-\mathcal{F}(\mathcal{D}(\mathbf{v})))^T\mathbf{C}_d^{-1}(\mathbf{d}-\mathcal{F}(\mathcal{D}(\mathbf{v})))\Bigg\}. \end{aligned} \tag{A-8}$$

In the optimization, maximizing $\sigma_v(\mathbf{v}|\mathbf{d})$ is equivalent to minimizing the following equation:

$$\mathcal{L}(\mathbf{v}) = \left\|\mathbf{C}_d^{-\frac{1}{2}}(\mathbf{d}-\mathcal{F}\circ\mathcal{D}(\mathbf{v}))\right\|^2 + \|\mathbf{v}\|^2. \tag{A-9}$$

Equation A-9 is a typical objective function including data misfit and L_2-norm regularization; thus it can be solved with the classical deterministic approach as elaborated in Habashy and Abubakar (2004).

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Biographies and photographs of the authors are not available.